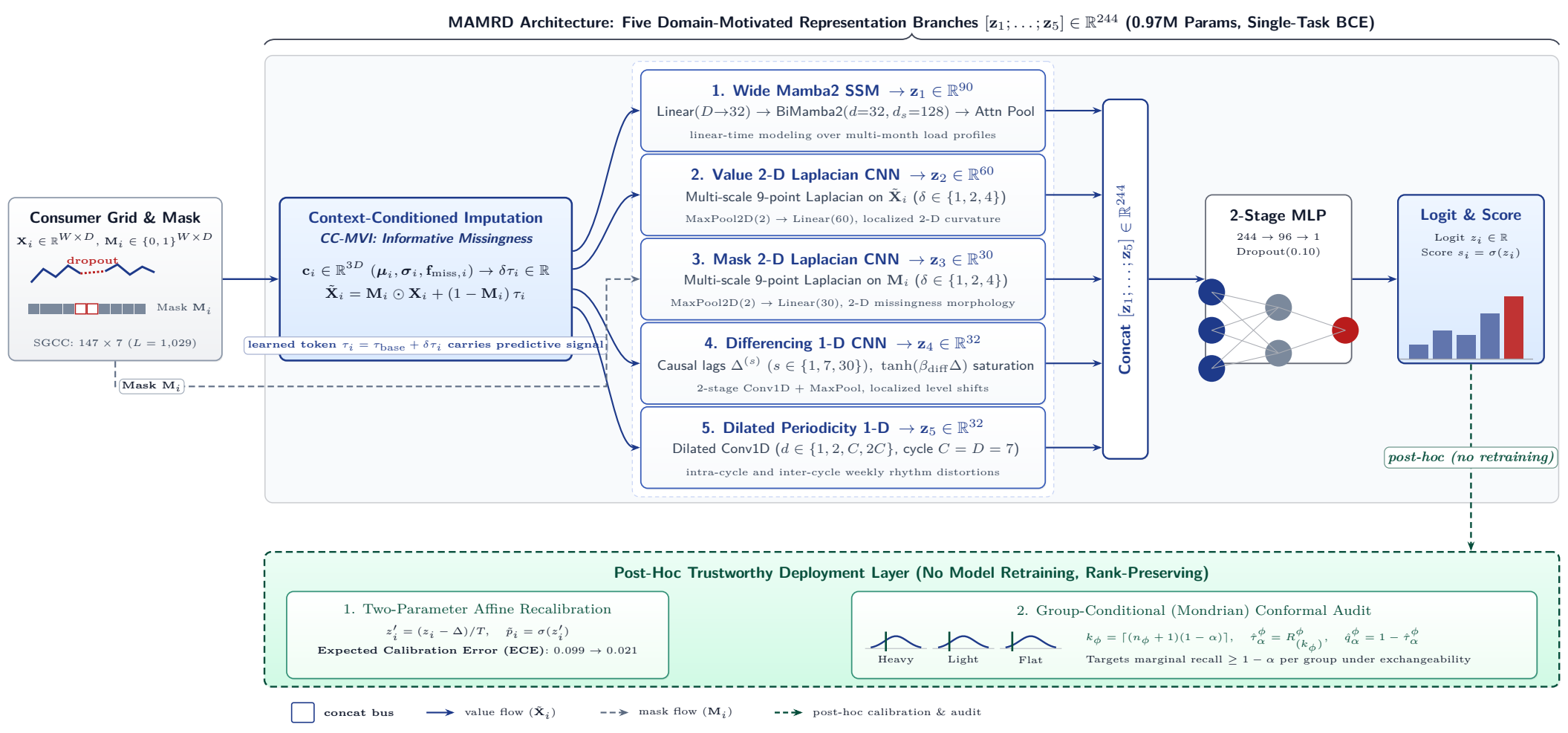

MAMRD Architecture: Five Domain-Motivated Representation Branches $[\mathbf{z}_1;\ldots;\mathbf{z}_5] \in \mathbb{R}^{244}$ (0.97M Params, Single-Task BCE)
Consumer Grid & Mask
$\mathbf{X}_i \in \mathbb{R}^{W\times D}$, $\mathbf{M}_i \in \{0,1\}^{W\times D}$
dropout
Mask $\mathbf{M}_i$
SGCC: 147 × 7 (L = 1,029)
Mask $\mathbf{M}_i$
Context-Conditioned Imputation
CC-MVI: Informative Missingness
$\mathbf{c}_i \in \mathbb{R}^{3D}$ $(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i, \mathbf{f}_{\text{miss},i}) \to \delta\tau_i \in \mathbb{R}$
$\tilde{\mathbf{X}}_i = \mathbf{M}_i \odot \mathbf{X}_i + (1-\mathbf{M}_i)\,\tau_i$
learned token $\tau_i = \tau_{\text{base}} + \delta\tau_i$ carries predictive signal
1. Wide Mamba2 SSM $\to \mathbf{z}_1 \in \mathbb{R}^{90}$
Linear(D→32) → BiMamba2(d=32, d_s=128) → Attn Pool
linear-time modeling over multi-month load profiles
2. Value 2-D Laplacian CNN $\to \mathbf{z}_2 \in \mathbb{R}^{60}$
Multi-scale 9-point Laplacian on $\tilde{\mathbf{X}}_i$ ($\delta \in \{1,2,4\}$)
MaxPool2D(2) → Linear(60), localized 2-D curvature
3. Mask 2-D Laplacian CNN $\to \mathbf{z}_3 \in \mathbb{R}^{30}$
Multi-scale 9-point Laplacian on $\mathbf{M}_i$ ($\delta \in \{1,2,4\}$)
MaxPool2D(2) → Linear(30), 2-D missingness morphology
4. Differencing 1-D CNN $\to \mathbf{z}_4 \in \mathbb{R}^{32}$
Causal lags $\Delta^{(s)}$ ($s \in \{1,7,30\}$), $\tanh(\beta_{\text{diff}}\Delta)$ saturation
2-stage Conv1D + MaxPool, localized level shifts
5. Dilated Periodicity 1-D $\to \mathbf{z}_5 \in \mathbb{R}^{32}$
Dilated Conv1D ($d \in \{1,2,C,2C\}$, cycle $C = D = 7$)
intra-cycle and inter-cycle weekly rhythm distortions
Concat $[\mathbf{z}_1;\ldots;\mathbf{z}_5] \in \mathbb{R}^{244}$
2-Stage MLP
244 → 96 → 1
Dropout(0.10)
Logit & Score
Logit $z_i \in \mathbb{R}$
Score $s_i = \sigma(z_i)$
post-hoc (no retraining)
Post-Hoc Trustworthy Deployment Layer (No Model Retraining, Rank-Preserving)
1. Two-Parameter Affine Recalibration
$z_i' = (z_i - \Delta)/T, \quad \tilde{p}_i = \sigma(z_i')$
Expected Calibration Error (ECE): 0.099 → 0.021
2. Group-Conditional (Mondrian) Conformal Audit
Heavy
Light
Flat
$k_\phi = \lceil (n_\phi + 1)(1-\alpha) \rceil, \quad \hat{\tau}_\alpha^\phi = R_{(k_\phi)}^\phi, \quad \hat{q}_\alpha^\phi = 1 - \hat{\tau}_\alpha^\phi$
Targets marginal recall ≥ $1-\alpha$ per group under exchangeability
concat bus
value flow ($\tilde{\mathbf{X}}_i$)
mask flow ($\mathbf{M}_i$)
post-hoc calibration & audit